

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01

Total Marks:100

Annexure A CASE STUDY

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 1: E-Commerce Customer Analysis

Scenario:

An e-commerce company has collected customer transaction data, including purchase amount, number of orders, browsing time, and customer ratings. The company wants to understand customer purchasing behavior and improve its marketing strategies.

Questions:

- A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize customer transaction data?
- B. Discuss how unusually high or low purchase amounts can affect the analysis and describe suitable methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether premium members spend significantly more than regular customers? Formulate the hypotheses and explain the analysis process.

Case Study 2: Renewable Energy Production Analysis

Scenario:

A solar power company has collected data from multiple solar plants, including daily energy generation, sunlight hours, panel temperature, and system efficiency. The company wants to evaluate plant performance and verify whether the average energy production meets the expected target.

Questions:

- A. Explain how descriptive statistics and distribution analysis can be used to summarize and evaluate the energy production dataset.
- B. Discuss the impact of abnormal energy generation values on the analysis and describe methods for detecting and handling outliers.
- C. How can confidence intervals and hypothesis testing be used to determine whether the average daily energy generation differs significantly from the expected target value?

Case Study 3: Hospital Patient Recovery Analysis

Scenario:

A hospital has collected patient recovery data, including age, treatment cost, number of hospital days, recovery time, and patient satisfaction score. The hospital wants to analyze treatment effectiveness and improve patient care quality.

Questions:

- A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize patient recovery time and treatment cost?
- B. Discuss how unusually high hospital stay duration or treatment cost values may affect the analysis. What methods can be used to detect outliers?
- C. How can hypothesis testing be used to determine whether patients receiving a new treatment recover faster than patients receiving the existing treatment? Formulate the hypotheses and explain the analysis process.

Case Study 4: Student Academic Performance Analysis

Scenario:

A university has collected student performance data, including attendance percentage,

assignment marks, internal test scores, study hours, and final exam marks. The university wants to identify factors affecting student success.

Questions:

- A. Explain how descriptive statistics and distribution analysis can be used to summarize student marks and attendance data.
- B. Discuss how extremely low or high marks can affect the interpretation of academic performance. Describe suitable methods for detecting and handling outliers.
- C. How can hypothesis testing be used to check whether students with attendance above 75% score significantly higher than students with attendance below 75%? Formulate the hypotheses and explain the steps.

Case Study 5: Banking Loan Approval Analysis

Scenario:

A bank has collected customer loan application data, including income, credit score, loan amount, repayment history, employment status, and loan approval status. The bank wants to understand customer risk and improve loan approval decisions.

Questions:

- A. How can descriptive statistics be used to summarize customer income, credit scores, and loan amounts?
- B. Discuss how unusually high loan amounts or very low credit scores can influence the analysis. Explain methods for identifying such outliers.
- C. How can hypothesis testing be used to determine whether customers with higher credit scores are more likely to receive loan approval than customers with lower credit scores? Formulate the hypotheses and describe the analysis process.

Case Study 6: Online Learning Platform Analysis

Scenario:

An online learning platform has collected learner activity data, including course completion percentage, quiz scores, video watch time, login frequency, and feedback ratings. The platform wants to improve course engagement and learner success.

Questions:

- A. Explain how descriptive statistics can be used to summarize quiz scores, watch time, and course completion rates.
- B. Discuss how unusually high or low learning activity values may affect the analysis. What techniques can be used to detect outliers?
- C. How can hypothesis testing be used to determine whether learners who watch more than 70% of the course videos score significantly higher in quizzes than other learners? Formulate the hypotheses and explain the testing procedure.

Case Study 7: Retail Sales Performance Analysis

Scenario:

A retail chain has collected sales data from different branches, including daily sales amount, number of customers, discount percentage, product category, and customer feedback score. The company wants to evaluate branch performance and improve sales planning.

Questions:

- A. How can descriptive statistics and distribution analysis be used to summarize daily sales and customer count?
- B. Discuss how abnormal sales values, such as festival-season sales or very low sales days, can affect the analysis. Explain suitable outlier detection methods.

C. How can confidence intervals and hypothesis testing be used to determine whether the average sales of urban branches are significantly higher than rural branches?

Case Study 8: Food Delivery Service Analysis

Scenario:

A food delivery company has collected order data, including delivery time, order value, customer rating, distance travelled, and delivery partner experience. The company wants to reduce delivery delays and improve customer satisfaction.

Questions:

A. Explain how descriptive statistics such as mean, median, standard deviation, and quartiles can be used to analyze delivery time and customer ratings.

B. Discuss how unusually long delivery times can affect customer satisfaction analysis. Describe suitable methods for detecting outliers.

C. How can hypothesis testing be used to determine whether experienced delivery partners complete deliveries faster than new delivery partners? Formulate the hypotheses and explain the analysis process.

Case Study 9: Manufacturing Defect Analysis

Scenario:

A manufacturing company has collected production data, including number of units produced, defect count, machine temperature, production time, and operator shift. The company wants to reduce defects and improve product quality.

Questions:

A. How can descriptive statistics be used to summarize defect counts, production time, and machine temperature?

B. Discuss how abnormal defect values may affect the quality analysis. What methods can be used to detect outliers in production data?

C. How can hypothesis testing be used to determine whether the defect rate differs significantly between day shift and night shift production? Formulate the hypotheses and explain the analysis process.

Case Study 10: Mobile Network Performance Analysis

Scenario:

A telecom company has collected mobile network data, including signal strength, internet speed, call drop rate, number of users, and customer complaints from different locations. The company wants to improve network quality.

Questions:

A. Explain how descriptive statistics and distribution analysis can be used to summarize internet speed, signal strength, and call drop rate.

B. Discuss how extremely low internet speed or high call drop rate values can affect the analysis. Describe suitable outlier detection and handling methods.

C. How can confidence intervals and hypothesis testing be used to determine whether the average internet speed in urban areas is significantly different from rural areas?

Case Study 11: Public Transportation Usage Analysis

Scenario:

A city transport department has collected bus usage data, including passenger count, ticket revenue, waiting time, travel duration, and route distance. The department wants to improve transport scheduling and passenger service.

Questions:

- A. How can descriptive statistics be used to summarize passenger count, waiting time, and travel duration?
- B. Discuss how unusually high passenger counts during peak hours or unusually long waiting times can affect the analysis. Explain methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether average waiting time during peak hours is significantly higher than during non-peak hours? Formulate the hypotheses and explain the testing process.

Case Study 12: Agriculture Crop Yield Analysis**Scenario:**

An agriculture department has collected crop production data from different farms, including rainfall, fertilizer usage, soil moisture, temperature, and crop yield. The department wants to analyze factors affecting crop productivity.

Questions:

- A. Explain how descriptive statistics and distribution analysis can be used to summarize crop yield, rainfall, and fertilizer usage.
- B. Discuss how abnormal crop yield values caused by drought, pests, or excessive rainfall can affect the analysis. Describe suitable outlier detection methods.
- C. How can confidence intervals and hypothesis testing be used to determine whether farms using organic fertilizer have significantly different crop yield compared with farms using chemical fertilizer?

Case Study 13: Employee Productivity Analysis**Scenario:**

A software company has collected employee productivity data, including working hours, number of completed tasks, project deadlines, experience level, and performance rating. The company wants to improve employee performance and project planning.

Questions:

- A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize employee productivity data?
- B. Discuss how unusually high or low task completion values can affect productivity analysis. Describe suitable methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether experienced employees complete significantly more tasks than newly joined employees? Formulate the hypotheses and explain the analysis process.

Case Study 14: Automobile Fuel Efficiency Analysis**Scenario:**

An automobile company has collected vehicle performance data, including fuel consumption, engine capacity, vehicle weight, mileage, and maintenance cost. The company wants to analyze fuel efficiency and improve vehicle design.

Questions:

- A. Explain how descriptive statistics and distribution analysis can be used to summarize mileage and fuel consumption data.
- B. Discuss how abnormal mileage values can affect the analysis. What methods can be used to detect and handle outliers?
- C. How can confidence intervals and hypothesis testing be used to determine whether hybrid cars have significantly better mileage than petrol cars?

Case Study 15: Social Media Engagement Analysis

Scenario:

A digital marketing agency has collected social media data, including number of likes, shares, comments, followers, posting time, and engagement rate. The agency wants to improve content strategy and audience reach.

Questions:

- A. How can descriptive statistics be used to summarize likes, shares, comments, and engagement rate?
- B. Discuss how viral posts with extremely high engagement values may affect the analysis. Describe suitable outlier detection methods.
- C. How can hypothesis testing be used to determine whether video posts receive significantly higher engagement than image posts? Formulate the hypotheses and explain the testing procedure.

Case Study 16: Airline Passenger Satisfaction Analysis

Scenario:

An airline company has collected passenger feedback data, including ticket price, flight delay time, baggage waiting time, seat comfort rating, and overall satisfaction score. The airline wants to improve service quality.

Questions:

- A. Explain how descriptive statistics can be used to summarize delay time, baggage waiting time, and passenger satisfaction scores.
- B. Discuss how unusually long flight delays can affect the analysis of passenger satisfaction. What methods can be used to detect outliers?
- C. How can hypothesis testing be used to determine whether passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights?

Case Study 17: Smart City Traffic Analysis

Scenario:

A smart city traffic department has collected traffic data, including vehicle count, average speed, signal waiting time, accident count, and pollution level from different roads. The department wants to reduce congestion and improve road safety.

Questions:

- A. How can descriptive statistics and distribution analysis be used to summarize vehicle count, waiting time, and average speed?
- B. Discuss how abnormal traffic values during accidents, festivals, or roadblocks can affect the analysis. Describe suitable methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether average waiting time is significantly higher during peak hours than during non-peak hours?

Case Study 18: Hotel Booking and Customer Review Analysis

Scenario:

A hotel chain has collected booking data, including room price, length of stay, customer rating, booking cancellation status, and service feedback score. The hotel wants to improve customer satisfaction and pricing strategy.

Questions:

- A. How can descriptive statistics be used to summarize room price, length of stay, and customer ratings?
- B. Discuss how unusually high room prices or extremely low customer ratings can affect the analysis. Explain suitable methods for detecting outliers.
- C. How can hypothesis testing be used to determine whether customers who stay more than three days give significantly different ratings compared with customers who stay for one or two days?

Case Study 19: Fitness App User Health Analysis**Scenario:**

A fitness app has collected user health data, including daily steps, calories burned, workout duration, heart rate, sleep hours, and fitness goal completion. The app wants to study user activity patterns and improve health recommendations.

Questions:

- A. Explain how descriptive statistics can be used to summarize daily steps, calories burned, workout duration, and sleep hours.
- B. Discuss how unusually high workout duration or abnormal heart rate values can affect the health analysis. What methods can be used to detect outliers?
- C. How can hypothesis testing be used to determine whether users who sleep at least seven hours per day burn significantly more calories than users who sleep less than seven hours?

Case Study 20: Online Examination Performance Analysis**Scenario:**

An educational institution has collected online exam data, including exam score, time taken to complete the test, number of attempts, internet interruption count, and student feedback. The institution wants to evaluate student performance and improve the online assessment system.

Questions:

- A. How can descriptive statistics and distribution analysis be used to summarize exam scores, completion time, and number of attempts?
- B. Discuss how unusually short completion time or extremely high number of attempts can affect exam performance analysis. Describe suitable outlier detection methods.
- C. How can hypothesis testing be used to determine whether students with fewer internet interruptions score significantly higher than students with frequent interruptions?